CALVIN TOLBERT

School of Operations Research and Information Engineering · Cornell University
cmt238@cornell.edu · calvintolbert.github.io

EDUCATION

Cornell University **Ithaca, NY**

Ph.D., Operations Research and Information Engineering May 2028 (expected)

- Advisor: David Bindel
- Research interests: Bayesian optimization, sampling algorithms, numerical methods in operations research

Stanford University **Stanford, CA**

M.S., Management Science and Engineering June 2023

B.A.S., Classics, Mathematics (Minor in Economics) June 2021

ASSISTANTSHIPS

Graduate Research Assistant to Professor Peter Frazier May 2026-August 2026

- Worked on a project through the Office of Naval Research to develop an AI harness for making military logistical decisions and formulating optimization problems.
- Was responsible for reviewing existing literature on AI for optimization and emphasizing originality of the harness.
- Researched preference learning in the context of the tool.

Course Assistant to Professors Jack Muckstadt and Eric Gentsch May 2025-August 2025

- Integrated course materials from ORIE's flagship supply chain management course into a self-contained instructor's packet.
- Split the lectures and assignments into units by topic and by mathematical content to make the course customizable to environments where different levels of mathematical preparation are assumed.
- Helped modernize the spreadsheets from the assignments and the graphics in the lecture slides.

Graduate Research Assistant to Professor Christina Lee Yu January 2024-August 2024

- Worked on matrix completion using tools from linear algebra, electrical engineering, and mathematical statistics.
- Presented our lab's work on using network flow to model potential outcomes in a fixed effects model.
- Wrote Python code to implement different matrix completion algorithms.

PUBLICATIONS

Preprints

"Nuclear norm-regularized Bayesian matrix completion" 2026
Calvin Tolbert

"Acquisition function optimization is NP-hard, even with tuned hyperparameters" 2026
Calvin Tolbert

Works in Progress

"Trend-informed Bayesian optimization" 2026
Darian Nwankwo, **Calvin Tolbert**, David Bindel

"Global optimization of acquisition functions" 2026
Calvin Tolbert, Michael Czekanski, David Bindel

PRESENTATIONS & TALKS

Invited Talks

"Nuclear norm-regularized Bayesian matrix completion"

Mathematics and Statistics Colloquium, San José State University — October 14, 2026

Departmental & Group Presentations

"Nuclear norm-regularized Bayesian matrix completion"

Bindel Lab, Cornell University — September 11, 2026

ORIE Ph.D. Colloquium, Cornell University — March 26, 2026

Conference & Workshop Presentations

"Trend-informed Bayesian optimization"

Young Researchers Workshop, Ithaca, NY (poster) — October 9, 2025

NYC Ops Day, New York, NY — March 27, 2025

"Minimax optimal estimates of individual causal effects in panel data under heterogeneous two-way fixed effects models"

American Causal Inference Conference, Seattle, WA (poster) — May 15, 2024

TEACHING EXPERIENCE

Guest Lecturer

ORIE 5110: Engineering Economics and Strategic Decision-Making — September 9, 2026
"Project Analysis: Loans and Interest, NPV/Payback/IRR"

ORIE 4126/5126: Principles of Supply Chain Management — February 24, 2026
"Mathematical Analysis of Supply Chain Behavior: The Inevitable 'Bullwhip Effect'"

Teaching Assistant

Cornell University courses, by semester:

MATH 2940: Linear Algebra for Engineers — Fall 2026

ORIE 5110: Engineering Economics and Strategic Decision-Making — Fall 2026

ORIE 4126/5126: Principles of Supply Chain Management — Spring 2026

MATH 1920: Multivariable Calculus for Engineers — Fall 2025

ORIE 3741/5741: Learning with Big Messy Data — Spring 2025

ORIE 4350/5350: Introduction to Game Theory — Fall 2024

Stanford University courses, by quarter:

MS&E 230: Market Design for Engineers — Spring 2023

MS&E 241: Economic Analysis — Winter 2023

TECHNICAL SKILLS

Programming & Tools: Python, Julia, GitHub

Methods: Optimization, numerical methods, Markov chain Monte Carlo, causal inference

SERVICE & PROFESSIONAL ACTIVITIES

- Organizer, Ph.D. Visit Days, Cornell ORIE, 2026-present

REFERENCES (THESIS COMMITTEE)

Professor David Bindel (bindel@cs.cornell.edu)
Department of Computer Science, Cornell University
Professor Peter Frazier (pf98@cornell.edu)
School of Operations Research and Information Engineering, Cornell University
Professor Jamol Pender (jjp274@cornell.edu)
School of Operations Research and Information Engineering, Cornell University